An isomorphism of partial subsystems over elementary subsystems is partially elementary

Silvia Barbina, Enrique Casanovas

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Editorial note. The counted statement is the author's environment at source0.tex lines 1001-1002 and its proof environment opens at line 1003; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial note. r002 note: the retained May_26_2024_2 context refers to Definition def5_1, so r002 adds exactly that definition as a further context and changes nothing else.

Editorial scope. One original result of the pinned author manuscript, counted once: isomorphisms of partial subsystems over elementary subsystems are partially elementary (author label elementary).

Allowed standard prerequisites (editorial).

- Partial elementary map. A map between subsets is partial elementary when it preserves the truth of formulas on the tuples where it is defined.. It is the conclusion of the statement.

- Elementary subsystem. A is elementary in M when the formulas with parameters in A agree.. It is the hypothesis on both sides.
- Saturated elementary extension. Every structure has a saturated elementary extension of prescribed large cardinality.. It is the reduction used in the countable case.

Definition 0.1 For $A, B \in \mathcal{C}_{\text{un}}$,

$A \leq B \Leftrightarrow A \subseteq B$ and there is an HF-ordering of B where A is an initial segment.

When $A \leq B$, we say that A is **strong** in B.

Lemma 0.2 If $<$ is an HF-ordering of B and $A = \text{cl}_<(A) \subseteq B$, then $<$ is an HF-ordering of B over A.

Lemma 0.3 Let M be an unconfined STS. If $A \leq M$ (as in Definition 0.1), then $\langle A \rangle \leq M$ and there is an HF-ordering of $\langle A \rangle$ over A whose final segment $\langle A \rangle \smallsetminus A$ is well-ordered. If N is an unconfined STS and $B \leq N$ and $f : A \cong B$ is an isomorphism of partial STSs, then f can be extended to an isomorphism $f' : \langle A \rangle \cong \langle B \rangle$.

Theorem 0.4 Let M, N be uncountable saturated unconfined STSs. The set of isomorphisms between countable substructures $A \subseteq M$ and $B \subseteq N$ (in particular, A, B are STSs) such that $A \leq M$ and $B \leq N$ is a back-and-forth system between M and N.

Corollary 0.5 Let M, N be infinite unconfined STS. If $A \leq M$, $B \leq N$ and $f : A \to B$ is an isomorphism of partial STS, then f is a partial elementary map from M to N.

Proof. First consider the case where A and B are countable. By Lemma 0.3, $\langle A \rangle \leq M$, $\langle B \rangle \leq N$ and f extends to some isomorphism $f' : \langle A \rangle \cong \langle B \rangle$. We can find uncountable saturated (or special ω_1-saturated) elementary extensions $M' \succeq M$ and $N' \succeq N$. Notice that $A \leq M'$ and $B \leq N'$. By Theorem 0.4, f' belongs to a back-and-forth system between M' and N' and so it is a partial elementary map from M' to N' and also from M to N.

Consider now the general case. Fix an HF-ordering of M having A as an initial segment. Consider a finite set $A_0 \subseteq A$ and let $A'_0 = \text{cl}_<(A_0)$. Since A is an initial segment, $A'_0 \subseteq A$. By Lemma 0.2, we may assume that A'_0 is an initial segment of A. Then $A'_0 \leq A$ and $A'_0 \leq M$. Now order B in such a way that f is an order isomorphism. This gives an HF-ordering of B. Moreover, if $B_0 = f(A_0)$ and $B'_0 = f(A'_0)$, then $B_0 \subseteq B'_0$ and B'_0 is an initial segment of B. Hence $B'_0 \leq B$ and $B'_0 \leq N$. Since A'_0 and B'_0 are countable, by the initial case we know that $f \restriction A'_0$ is elementary. Hence, f is elementary. □